



Objective & Lecture Mapping: In previous labs, we built a Simple Reflex Agent that moved randomly or reacted only to immediate adjacent cells. Today, we transition to a **Goal-Based/Planning Agent**. You will expose the environment's state space to the agent, allowing it to use **Breadth-First Search (BFS)**, **Depth-First Search (DFS)**, and **Uniform-Cost Search (UCS)** to simulate and find paths to the food before taking a physical action.

Part 1: Practical Implementation — Building the Search Agent

Step 1.1: Exposing the World Model

Classical search algorithms require an abstract model of the environment to simulate future states. Currently, your agent is "blind" to the overall map.

1. Create a new Branch called Week_03 in your Forked Github Repo and work on the below Questions.
2. Open `visual_grid_game.py`.
3. Locate the `get_percept(self)` method.
4. Add three new keys to the dictionary to pass the global state to the agent:

```
'grid_size': (self.width, self.height)
'walls': list(self.walls)
'all_food': list(self.food_positions)
```

Step 1.2: Implementing BFS, DFS, and UCS

You will implement three distinct uninformed search strategies. The core node expansion structure is identical; only the data structure for the **Frontier** changes!

1. Open `agent.py` and create a new class called `SearchAgent` .
2. Import required structures: `from collections import deque` and `import heapq` .
3. **BFS**: Create `def bfs_search(...)` . Use a FIFO queue (`deque.popleft()`). This explores the shallowest nodes first.
4. **DFS**: Create `def dfs_search(...)` . Use a LIFO stack (`list.pop()`). This explores the deepest nodes first.
5. **UCS**: Create `def ucs_search(...)` . Use a Priority Queue (`heapq.heappop()`) ordered by the total path cost $g(n)$.
6. For all three algorithms, you must maintain a `reached` set (or visited array) to track explored states. This converts your algorithm from a *Tree Search* to a *Graph Search*, preventing infinite loops.

Step 1.3: Executing and Comparing Offline Plans

The agent must now form a complete plan and execute it step-by-step.

1. In your `SearchAgent.__init__` , add an empty list: `self.plan = []` and a config string `self.active_algo = 'BFS'` .
2. In `sense_and_act(self, percept)` , check if `self.plan` is empty.
3. If empty, find the closest food pellet from `percept['all_food']` , execute the search method matching `self.active_algo` , and store the resulting sequence of actions in `self.plan` .
4. Return the first action from the plan using `return self.plan.pop(0)` .
5. **Observation Task**: Change `self.active_algo` between 'BFS', 'DFS', and 'UCS' and run the simulation. Watch how DFS takes erratic, winding paths, while BFS and UCS take direct, optimal paths!

Part 2: Theoretical Evaluation

Based on your implementation and Lecture 04, complete the following analytical questions.

1. (Remember) You built your search logic based on the environment coordinates. What is the fundamental difference between the "State Space" and the "Search Tree"?

The State Space is the complete set of all possible states or positions that can exist in the environment, while the Search Tree is the structure created by the search algorithm as it explores those states from the initial state.

In our grid, the state space contains all valid grid positions, while the search tree contains the paths and nodes explored by BFS, DFS, or UCS to reach the goal.

2. (Understand) In Step 1.2, you were instructed to use a `reached` set. In graph search theory, what specific problem does this set solve, and what would happen to your DFS agent without it?

The 'reached' set prevents the agent from revisiting states it has already explored, avoiding loops and repeated work. Without it, DFS could continuously move between the same states and potentially run forever.

3. (Analyze) Compare the visual paths generated by BFS and DFS in your simulation. Why is BFS guaranteed to be optimal in this specific grid, whereas DFS produces winding, suboptimal routes?

BFS explores the grid level by level, so it always finds the path with the fewest steps. Since every movement in this grid has the same cost, BFS is guaranteed to find an optimal path, while DFS explores deeply in one direction first and can therefore produce longer, winding paths.

4. (Evaluate) If we scaled `visual\_grid\_game.py` up to a massive 1000x1000 grid, BFS and UCS might crash before finding food. According to the time and space complexity formulas from the lecture, contrast the memory bottlenecks of BFS versus DFS.

BFS has a space complexity of $O(b^d)$, so its memory usage grows very quickly as the solution depth increases, making it the major memory bottleneck on a 1000x1000 grid. DFS uses only $O(bm)$ space, so it requires much less memory, although it may take longer to find the food and does not guarantee an optimal path.

Once Completed Get a PDF of the completed File and Push into the Week 03 brach in the Github Project

Finalize and Lock Lab 03 Documentation



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